Deep Learning based Bone Fracture Detection

Sailaja Thota
School of CSE
REVA University
Bangalore, 560064, India
sailaja.thota@reva.edu.in

Pranav Kandukuru
Department of ECE
NIT Calicut
Kattangal, Kerala 673601, India
Pranav_b210612ec@nitc.ac.in

Meenakshi Sundaram
School of CSE
REVA University
Bangalore, 560064, India
meenakshi.sa@reva.edu.in

Anooja Ali
School of CSE
REVA University
Bangalore, 560064, India
anooja.ali@reva.edu.in

Syed Muzamil Basha
School of CSE
REVA University
Bangalore, 560064, India
muzamilbasha.s@reva.edu.in

Hima Bindu N
School of EEE
REVA University
Bangalore, 560064, India
hima.bindu@reva.edu.in

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***Abstract*— A current trend across several industries involves utilizing computer-based technologies to identify faults. To meet the demands of immediate detection and high precision, a highly responsive system should leverage modern approaches and make full use of available resources. While various methods exist for detecting bone fractures in the modern world, such as Magnetic Resonance Imaging (MRI), CT scans, and Bone scans, these approaches tend to be more expensive, uncomfortable for patients, and less effective at detecting subtle fractures that, if left untreated, could lead to significant challenges. In recent years, the application of Convolutional Neural Networks (CNNs) in medical image fracture identification has shown promise in automating the detection of bone fractures from X-ray images. However, deploying such algorithms on devices remains challenging due to limited computing resources. In this research work, MobileNet, employs X-ray images to detect bone fractures, and its results are compared with those of a CNN model. The MobileNet architecture is chosen for its capacity to reduce computational complexity while maintaining high accuracy of 98%.**

Keywords*— *CNN algorithm, MobileNet, fracture detection, deep learning model.

I. INTRODUCTION

X-ray imaging is a rapid and non-invasive technique, used as a primary diagnostic tool used to confirm or rule out the presence of fractures. The combination of radiation and computer image processing technology has led to the widespread usage of digital X-ray imaging equipment in several medical applications[1]. By visually examining an X-ray image and applying appropriate therapies, it is common practice to assess the existence and severity of an injury. X-ray images offer high-quality and detailed visualizations of bone structures can help identify potential complications associated with fractures, such as displaced bone fragments, joint involvement, or damage to surrounding tissues. The potential risks associated with radiation exposure need to be carefully considered, especially when repeated X-rays are required.

Bone fracture is a common injury affecting millions of people worldwide. Accurate and timely diagnosis of bone fractures is essential for effective treatment and patient recovery. Bone fractures can be partial or complete type and open or closed type according to the exposure of skin. Fractures can also arise from conditions that weaken bone integrity, like osteoporosis or bone cancer [2]. The indications of a bone fracture may encompass abrupt pain, noticeable misalignment, bruising, swelling, as well as a sensation of warmth or redness in the affected area. Although studies of gene expression in endochondral bone repair have found that multiple components of the extracellular matrix and growth factor of genes that are important in the growth of tissues expressed during the various stages of fracture repair, the molecular pathways that regulate bone repair are still largely unknown.

For asymptomatic diagnosis of bone fractures, X-ray imaging is widely used. Medical experts may find it troublesome to interpret X-ray visuals, and a false positive diagnosis might result to incorrect remedy and a protracted recovery. Recent developments in computer vision and algorithmic learning have showed significant potential in automating the diagnosis and identification of fractured bones from X-ray pictures. Due to the complexity of the human skeleton, sophisticated algorithms are required to analyses and spot abnormalities in medical imaging.

In machine learning, the features are not automatically chosen; instead, the user must process the maximum number of features, which is challenging if the input is an image dataset [3]. Researchers have evaluated deep learning techniques, particularly CNN and MobileNet models have demonstrated remarkable capabilities in image recognition tasks, including fracture detection in large datasets of labelled X-ray images to detect and classify fractures with high accuracy [4]. These models learn to identify patterns and features associated with different types of fractures. Deep learning models simplify human feature selection and address the difficulty of the issue by partitioning it into many mappings are suggested as solutions to these kinds of problems. This is broken into a number of nested basic mappings with various model layers rather than a single complicated mapping. The input layer has received the raw

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data, and as more and more hidden layers extract abstract features from the image for which prior values were not provided, the models themselves must choose the concepts that will be used to elucidate how the observed facts are connected [5].

In this study, a novel method for identifying bone fractures has been put forth and it directs in creating a smart classification system with the ability to detect and categorize bone fractures. An architecture called MobileNet is suggested, which analyses X-ray images and looks for bone fractures. The MobileNet architecture is used to reduce the computational complexity of the CNN model, making it suitable for real-time fracture detection applications. With the use of a large collection of X-ray imagery from Kaggle [6], the algorithm is trained and evaluated.

II. Literature Survey

The approaches for recognizing and categorizing fractures involve data preparation, image pre-processing, component extraction, and grouping. Expert radiologists are required to examine and classify radiographic images, a process that is both tedious and costly. This procedure could potentially be simplified by implementing fracture classification algorithms. According to the author [7], the scarcity of labelled training data sets is the most critical obstacle in constructing a high-performance classification model. The goal is to present a variety of concepts and perspectives that may aid in the development of an accurate model capable of identifying multiple types of fractures across various anatomical sites.

In [8], a novel 3-D annotation approach for fractures, referred to as '3-D surface annotation,' was introduced. This approach significantly reduces the annotation burden by utilizing 3-D shapes of pelvic surfaces to calculate fracture intensity at each spot within the pelvic region. The evaluation involves analyzing each triangle within the 3-D data, with identification classes categorized into two groups: the fracture class and the non-fracture class. The accuracy, precision, recall, and specificity of the test data were found to be 69.5%, 61.1%, 56.4%, and 7.7%, respectively.

The authors of [9] employed the CNN Algorithm, which classifies data into two categories: normal and abnormal. In the typical scenario, the algorithm bypasses the segmentation step and displays 'not impacted,' leading to the termination of the procedure. However, if the case is deemed atypical, further classification is conducted. The Bone Fracture Detection System underwent testing on a distinct dataset of X-ray images, achieving an accuracy of 78%.

The authors of [10] proposed an extended deep CNN architecture, which represents a broader version of the widely utilized U-Net model. This CNN produces two distinct outputs: the probability of a radiograph containing a visible fracture and a heat map that indicates the likelihood of a fracture's presence at each point within the image. The training process of the model can be divided into two main steps. Initially, the model underwent pre-training using a designated pre-training dataset. Subsequently, the acquired model was fine-tuned during the second step using a specialized training dataset to enhance its proficiency in identifying and localizing wrist fractures. To mitigate over-fitting, the authors employed an array of techniques, including early stopping, data augmentation, and dropout, with an estimated probability of 0.5.

A unique approach to vertebral compression fracture (VCF) detection, based on 3D localization and combining Deep Reinforcement Learning and Imitation Learning, is detailed in [11]. While some studies include 3D bounding box localization in CT images as a processing step, many focus on pinpointing the Region of Interest's central point, along with reporting the average separation between projected and actual centers. Among the works utilizing bounding box localization and providing Jacquard Index/Dice Coefficient after final segmentation on a limited dataset, comparing performance solely on localization becomes challenging. To identify VCF, the localized area undergoes analysis using a patch-based CNN. The normalized accuracy of the 6-layered CNN is 80% when evaluated on a dataset of 127 CT examinations.

Using X-ray images, the author of [12] proposed a system for detecting fractures in the seven upper extremity bones: shoulder, humerus, forearm, elbow, wrist, hand, and finger. The system employed a two-stage classification algorithm built upon the MobileNet network framework. In the first stage, an enhanced X-ray image is utilized to determine the bone type. Subsequently, based on the outcome of the first stage, the bone image was fed into one of the seven classification systems (one for each bone type) to detect anomalies within the bone. Upon combining the results of these two classification stages, the average accuracy achieved was 73.42% using the MURA dataset as the benchmark performance dataset.

In the study of [13], 3392 X-ray radiographs of arm bone fractures from the MURA dataset were employed to train the algorithms, while the remaining 612 radiographs were used as a test set to evaluate the method. The bounding boxes for the arm fracture X-ray radiographs were annotated by qualified radiologists. The study's results demonstrated that, even with the presence of low-quality images within the MURA dataset, the proposed technique achieved a mean accuracy of 62.04% in identifying arm fractures, surpassing other state-of-the-art deep learning methods.

The author of [14] proposed a method that not only detects bone fractures but also precisely locates them in an X-ray image, potentially aiding doctors in faster fracture detection. The Crack-Sensitive CNN approach surpasses the method that utilizes the Schmid filter for image pre-processing followed by ResNet for classification. Among the various networks compared, Crack Net exhibits the highest recall, surpassing the others by 3%. Consequently, the Schmid convolutional layer appears to be more sensitive to fractures. In conclusion, Crack Net is better suited for bone fracture imaging.

The author of [15] introduced a deep learning approach utilizing X-ray images for the early diagnosis of bone disorders and the identification of different bone fractures. The initial stage involved pre-processing, encompassing normalization and augmentation. Subsequently, classification was conducted by tuning the parameters of the CNN model. To determine the most suitable model, several crucial criteria

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were examined, including the number of epochs, batch size, type of optimizers, and learning rate. The CNN model exhibited strong performance, achieving a specificity of 89.865%, an accuracy of approximately 90%, and an area under the Receiver Operating Characteristic curve of 0.8088.

The proposed system of [16] consists of two main phases. The initial phase involves processing fracture images using various image processing techniques to identify their locations and shapes. The subsequent phase is the classification step, wherein a back propagation neural network is trained and subsequently tested on the processed images. Through experimentation, the system was assessed using diverse bone fracture images, yielding impressive efficiency and a commendable classification rate.

Previous studies have limitations in comparing with other state-of-the-art methods or networks concerning performance metrics such as accuracy, sensitivity, specificity, or computational efficiency. Improving the analysis and comparison with existing techniques could validate the effectiveness of the proposed method. Moreover, the study may have used restricted evaluation metrics, impeding comprehensive analysis and comparison with existing approaches, thereby presenting a research gap. The majority of prior research focuses solely on a single bone, primarily due to the considerable variability in the shapes of different bone types and the limited availability of comprehensive datasets. In our work, we are addressing this gap by considering additional metrics such as precision and recall alongside accuracy.

III. METHODOLOGY

This proposed methodology introduces a method for automating bone fracture identification using CNN and MobileNet models. The method involves collecting a dataset of 5000 labelled Bone X-ray images and performing data preprocessing to enhance model robustness. The convolutional neural network (CNN) architecture encompasses layers for convolution, pooling, and full connection. In contrast, the MobileNet model is a resource-efficient blueprint that leverages depth-wise separable convolutions to craft condensed yet intricate convolutional neural networks. This architecture is particularly well-suited for mobile and embedded vision tasks, offering a resource-conscious solution. These models, designed to operate under limited power, can be customized to suit a range of needs such as classification, detection, embeddings, and segmentation. MobileNet finds extensive application in various practical scenarios, including object detection, detailed categorization, facial attribute analysis, and localization.

Depth-wise separable convolution, utilized CNN to process grid-like data like images, aims to streamline computational demands while upholding accuracy across tasks such as image classification, object detection, and semantic segmentation.

In the context of the traditional convolutional approach, a solitary filter, also referred to as a kernel, traverses across all input channels within the input feature map. Consequently, the same filter is responsible for extracting features from each input channel. This procedure's computations can be resource-intensive, particularly when handling extensive input channels and filters.

In contrast, Depth-wise separable convolution disassembles the convolution operation into two distinct stages: depth-wise convolution and point-wise convolution.

- Depth-wise Convolution:

During the depth-wise convolution phase, individual input channels undergo separate convolution with their dedicated set of filters. Each input channel is essentially treated with its own filter. This step enables the independent capture of spatial features across input channels.

- Point-wise Convolution:

Following the depth-wise convolution, a point-wise convolution (sometimes referred to as 1x1 convolution) is implemented. In this step, a 1x1 filter is utilized to linearly amalgamate outputs derived from the depth-wise convolution. This process facilitates the creation of novel features by intermixing the features extracted from diverse input channels.

Advantages of depth-wise separable Convolution:

- Diminished Computational Complexity: By segmenting the convolution into distinct depth-wise and point-wise stages, the volume of computations is significantly curtailed. This is especially advantageous in terms of economizing memory and computation resources.
- Reduced Parameter Count: Depth-wise separable convolutions typically necessitate fewer parameters compared to conventional convolutions. This can culminate in a more streamlined model architecture, particularly in contexts with restricted computational capabilities.
- Enhanced Generalization: Depth-wise separable convolution enables the network to learn distinct and diverse features, potentially augmenting its ability to generalize across various tasks.
- Regularization: The separable characteristic of this convolution functions as a form of regularization, assisting in averting over-fitting to the training data.
- Multi-scale Representations: Depth-wise separable convolutions can be employed to establish multi-scale representations by stacking layers with varying kernel sizes.

The trained models are optimized through experimentation and deployed using TensorFlow and Django frameworks. The models classify X-ray images as Fracture or Non-Fracture based on evaluation metrics, ensuring accurate fracture detection. The system includes dataset collection, preprocessing, training, deployment, and output classification components, working together to automate and improve bone fracture identification. The process flow for CNN/MobileNet model is shown below in Fig 1.

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3.1 Data Collection

The collected dataset for CNN model and MobileNet model contains 5000 Bone X-ray images from Kaggle [6]. The photos have been labelled with the appropriate class, such as fractures and non-fractures. It is separated between sets of testing and training with a 70:30 ratio.

3.2 Data Pre-processing

Pre-processing of the Bone X-ray pictures includes standardizing them to a consistent size of 256x256 and normalizing the pixel values to the range [0, 1]. Data augmentation methods such as rescaling are used to increase the size of the training set, and ImageDataGenerator is used to improve the robustness of the model [17]. This class enables the model to receive a fresh variant of the pictures at each epoch, increasing the size of the training set and improving the model's robustness.

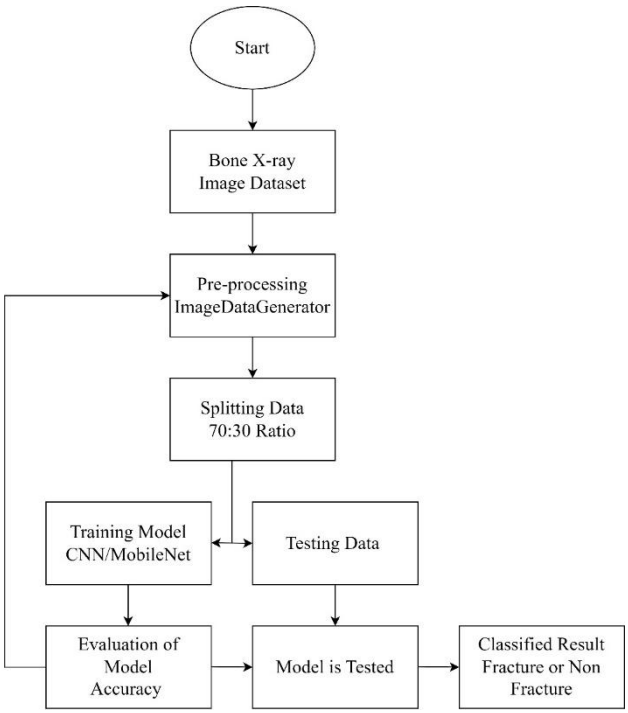

Fig 1: Process flowchart of CNN/MobileNet model

3.3 Model Selection

A deep learning CNN is composed of three layers: convolutional, pooling, and fully connected (FC). A large number of calculations take place in the convolutional layer, which is the foundation of a CNN. The pooling layer minimises the amount of parameters in the input and reduces elimination of some data. The Fully Connected layer in the CNN is where picture categorization occurs based on the characteristics extracted in the preceding layers. Fully connected in this context indicates that all of the inputs or nodes from one layer have links to every activation unit or node in the following layer.

MobileNet is a lightweight CNN architecture that reduces the amount of components and calculations required through using depth-wise separable convolutions. The pre-trained MobileNet model serves as the starting point for our bone fracture detection model on the dataset.

3.4 Training the Model

For the input and hidden layers of the CNN model, ReLU was chosen as the activation function, while Softmax was employed for the output layer. Adam optimizer, batch size, categorical cross entropy loss function, and 20 epochs were utilized for maximum precision.

The MobileNet model is customized with ReLU activation functions for the input and hidden layers, and sigmoid activation functions for the output layer. For good efficiency, the cross-entropy loss function, the Adam optimizer, and a grid search were used to optimize the necessary parameters, such as batch size and number of epochs selected.

DenseNet utilizes dense blocks as fundamental module units, where each dense block incorporates four closely connected layers featuring a growth rate of 4. Within this configuration, every layer uses the output feature maps from preceding layers as its input feature maps.

The dense block simply stacks the feature maps from preceding convolutional layers and augments the count of feature maps. As a result, it's only necessary for the magnitudes of and to match, while the quantity of feature maps isn't necessarily uniform is depicted in Fig 2. DenseNet employs the growth rate hyperparameter to manage the quantity of feature map channels in the network. The growth rate signifies that each network layer's output feature maps consist of additional channels. In essence, for every convolutional layer, the subsequent layer's input feature maps witness an augmentation of channels.

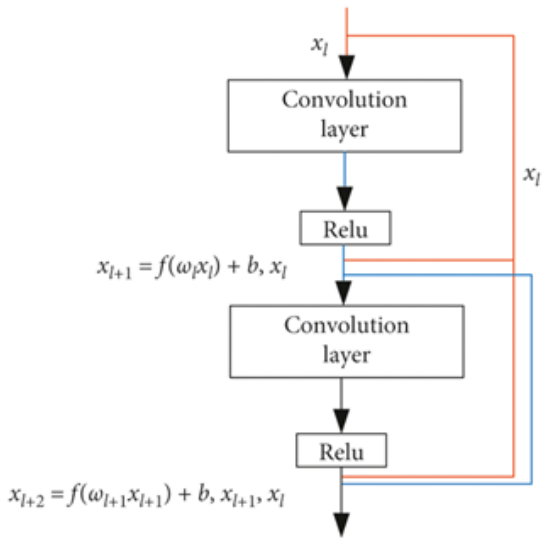

Fig 2: Two convolution layers among dense connected modules

3.5 Model Evaluation

Evaluation of the trained model on the testing set using metrics such as accuracy for checking the performance.

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3.6 Model Optimization

Experimentation with different dropout rates, regularization techniques, and activation functions will help for model optimization and also improve the performance of the model.

3.7 Classification Model

The trained model will be deployed on a device using TensorFlow and Django, a framework for running machine learning models on embedded devices. The performance of the model on a device is a test to find whether it will process an X-ray image in less time with a high accuracy [18]. The output is classified as Fracture or Non Fracture along and consider accuracy as an evaluation metrics.

IV. RESULTS AND DISCUSSION

The MobileNet model was trained on a huge collection of X-ray images with and without fractures, and its performance on the test dataset was compared to CNN. The user interface was created with Python and Django, as seen in Fig 3. Using the developed web application, the user may submit an input X-ray picture and select either the MobileNet Model or the CNN Model, as shown in Fig 4a and 4b respectively, to determine if the image is fractured or not.

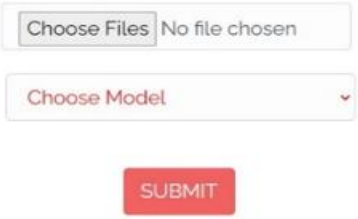


Fig 3: Web Application Interface

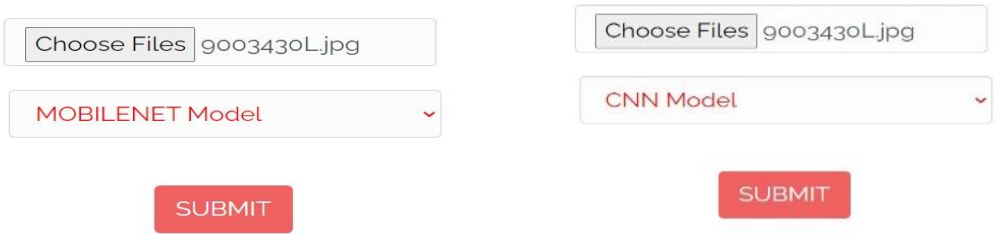
Fig 5a shows the result of an X-ray image which was uploaded and classified as Non-Fracture and Fig 5b classified as Fracture using MobileNet model. MobileNet architectures feature a reduced parameter count compared to traditional CNNs, leading to decreased overfitting risks. Their reduced computational demands, owing to fewer computations, make them a viable option for mobile vision applications.

Fig 6a shows the result of an X-ray image which was uploaded and classified as Non-Fracture and Fig 6b classified as Fracture using CNN model.

The bone fracture detection system was evaluated for a distinct set of X-ray pictures and found that the system with the CNN model has an accuracy of 86% and the system with the MobileNet model has an accuracy of 98%, assisting in the efficient diagnosis of the existence of bone fracture as mentioned in Table 1.

Table 1: Comparison of performance measures

Model Evaluated	Types of Bone	Accuracy [19]	Precision	Recall
CNN	Fracture	85.956	89	91
	Non Fracture	85.956	92	83
MobileNet	Fracture	98.189	91	92
	Non Fracture	98.189	95	85



a) Upload Image and choosing MobileNet model

b) Upload Image and choosing CNN model

Fig 4: User interface to upload an image and choose the classifier model

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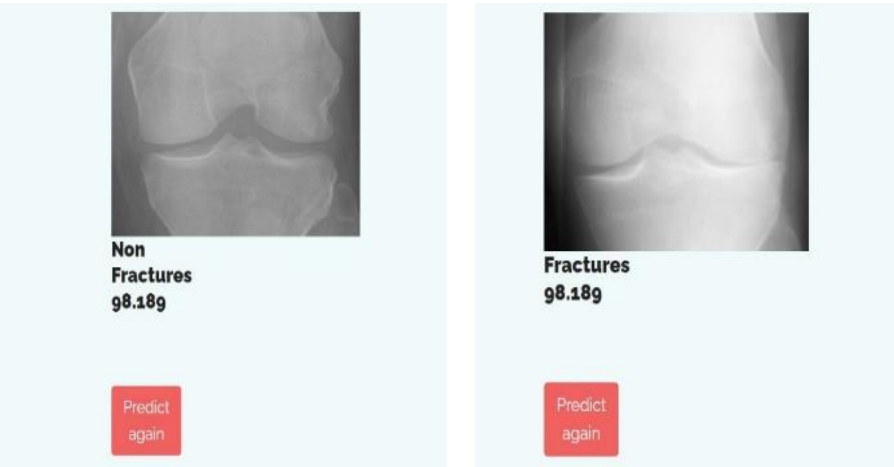
a) Non-Fracture Result b) Fracture Result

Fig 5: Result of MobileNet model



a) Non-Fracture Result b) Fracture Result

Fig 6: Result of CNN model

V. CONCLUSION

Anomalies within the bones are regarded as significant medical instances, as an incorrect diagnosis could result in adverse repercussions. As a result, computer-assisted diagnostic systems play a crucial role in contemporary healthcare. This research introduces an approach to identify bone fractures, suggesting a two-phase classification technique utilizing the MobileNet network. The trained model accurately detects bone fractures from X-ray images with a high accuracy and F1 score. The bone fracture detection system was evaluated on a separate dataset of X-ray pictures and discovered that the system with the CNN model has an accuracy of 86% and the system with the MobileNet model has an accuracy of 98%, which aids in determining whether there is a fracture in the bone or not in an efficient manner. The CNN model's computational cost is decreased by the MobileNet architecture, enabling real-time fracture detection applications. However, the current deployment of the model on mobile devices may encounter challenges in optimizing inference speed and ensuring compatibility with diverse hardware configurations. Therefore, efforts should be directed towards refining deployment strategies for mobile devices, optimizing inference speed, and ensuring compatibility with varying hardware configurations. These advancements would significantly enhance the practical utility and effectiveness of bone fracture detection systems in clinical settings.

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